Models, evidence and economic policy

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Economists use models and evidence to answer questions about policy. Would a tariff help consumers? How would leaving a trade agreement affect living standards? To assess an answer, you need to understand the comparison being made, the mechanism and the assumptions. You do not need to reproduce every equation in the paper.

Start with the comparison

A policy effect is the difference between what happens with the policy and what would have happened without it. That second outcome is the **counterfactual**. We observe only one of these paths, so the other must be estimated.

For example, comparing UK income today with UK income before Brexit does not isolate the effect of Brexit. Many other things changed over the same period. Income could grow after Brexit while remaining below the level it would have reached without Brexit.

Before a policy happens: use a model

A model is a simplified account of how an economy works. It links a policy change to people's and firms' decisions, and then to outcomes such as prices, trade and income. A trade diagram is already a model. A larger model adds more countries, industries and connections between them.

To turn a mechanism into a numerical prediction, economists combine data with assumptions about how strongly people and firms respond. Those responses may come from earlier empirical studies. They then compare outcomes under different policy arrangements. This is a **model simulation**.

Dhingra and colleagues used this approach in 2016, before Brexit. They compared remaining in the EU with different possible arrangements for leaving. Their results depended on assumptions about tariffs, other trade costs and future integration within the EU. A scenario is a conditional prediction: it describes what the model implies if those assumptions hold.

A simple mechanism: the cost of paperwork

Imagine that selling abroad requires a business to spend an extra £300 a year on paperwork, regardless of how much it sells. For a firm selling 10 units, that adds £30 per unit. For one selling 1,000 units, it adds 30p per unit. These are invented numbers, used only to illustrate a fixed cost.

The smaller exporter may find that trading is no longer worthwhile. This helps explain why removing tariffs need not remove every obstacle to trade, and why the same rule can affect firms differently. The example gives a possible explanation; evidence is needed to establish whether it matters in practice.

After a policy happens: look for a fair comparison

Once a policy is in place, researchers can study the data. But a change after the policy is not automatically a change caused by it. If prices rise, the cause might include energy costs, exchange rates or other events as well as the policy.

One approach compares changes for groups more exposed to the policy with changes for groups less exposed. Freeman and colleagues compare UK firms' trade with the EU and the rest of the world, while accounting for other influences. The basic question is whether trade changed differently where the new barriers applied.

Such a comparison still needs assumptions. Would the groups have changed similarly without the policy? Did another event affect them differently? Could the policy itself affect the comparison group? Researchers use the data and additional checks to assess these concerns. A strong correlation or a statistically significant result does not, on its own, establish cause and effect.

Models and evidence work together

Models suggest mechanisms and help organise the evidence. Empirical studies estimate responses, test explanations and reveal where a model misses something. Both approaches involve assumptions. A useful assessment explains which assumptions matter and how the conclusion might change if they fail.

The two Brexit readings do different jobs. Dhingra and colleagues predict changes in living standards under specified arrangements. Freeman and colleagues study firms' goods trade after the new arrangements began. Their findings can inform the same argument, but a percentage change in exports is not the same outcome as a percentage change in income.

Read the claim carefully

- **What is measured?** Trade, prices, profits, GDP and welfare are different outcomes. A finding about one does not automatically establish the effect on another.
- **Compared with what?** An economy being smaller than it otherwise would have been does not mean that it has shrunk compared with an earlier year.
- **Over what period?** An initial disruption, a lasting difference in income and a change in annual growth are different claims.
- **For whom?** An average can conceal substantial differences between firms or households.
- **How certain is the conclusion?** State the result alongside an assumption or limitation that matters for interpreting it.

When preparing an essay, connect the mechanism to a specific finding, explain why that evidence is relevant, and identify what it cannot establish. Describing a result is a starting point; explaining the comparison and its limits makes it an argument.

Sources and attribution

Adapted and rewritten by David Chivers from Sara Eugeni's introductory notes on model simulation and econometrics, October 2026. The numerical example above is illustrative.

The seminar readings are Dhingra et al. (2016), The consequences of Brexit for UK trade and living standards, and Freeman et al. (2025), How Brexit affected the trade of UK firms. The seminar sheet identifies the pages to read.